

Exercises

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Surname, First name

Applied Microeconomics (FEB12001X)
 FEB12001X - Midterm 17 September 2025

1	1	1	1	1	1
2	2	2	2	2	2
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General information

Lecturers: Josse Delfgaauw and Robert Dur

Date & Time of examination: 17 September 2025, 18:30 - 19:30

Type of examination: Closed book, Open questions

Number of questions: 3 questions (question 4 is additional answer space)

Number of pages: 10 pages (incl. cover sheet).

Separate appendix: no

Instructions

- Please use only a pen with black or blue ink.
- Fill in your name and student number as indicated above. **Also, code your student number by coloring the circles with the corresponding numbers.** If you made a mistake, put a cross through the wrong number, and color the correct one.
- Write your answer **in** the answer box corresponding to the question.
- **You have to provide a calculation or explanation of each answer.** An answer without calculation or explanation yields zero points.
- Because the total answer space is limited, we strongly advise you to prepare your answers on scrap paper: **no additional exam paper will be provided.**
- In case you made a mistake and therefore you need more space, use the additional writing space at the end of the exam. Clearly indicate where the (remaining part of the) answer can be found.

Permitted materials

- You are allowed to use a pen and blank scrap paper.
- You are allowed to use a calculator (only the following types: FX-82SX Plus / FX-82MS/NL / FX-350MS / FX-570MS / TI-30X:a/S/B/IIS/IIB).
- You are NOT allowed to use notes.
- You are NOT allowed to use books.
- You are NOT allowed to use a dictionary.

- the cover sheet continues on the next page -



Please note:

- Exam questions and/or answers may not be copied or disseminated in any way.
- Scrap paper and other exam related documents may not leave the exam room.
- It is not allowed to keep a watch within reach during the examination. Watches must be stored in your coat or bag.
- All mobile devices or (potential) data media must be switched off during the examination and stored in your coat or bag.

Good luck!

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Exercise 1 Public Economics

Consider two individuals, A and B, who both derive utility from consuming a pure public good. For individual A, the utility from consuming Q units of the public good (measured in euros) is given by

$$U^A = aQ - \frac{1}{2}Q^2$$

The utility of individual B from consuming Q units of the public good (measured in euros) is given by

$$U^B = bQ - \frac{1}{2}Q^2$$

The marginal production cost of the public good are equal to c euro, which is also the price per unit of the public good on the market. Assume that $a > b > c > 0$.

- 1p **1a** Determine the socially efficient level of provision of the public good

- 1p **1b** Suppose A and B decide independently how many units of the public good to buy (of course taking the presence of the other into account). Determine how many units person B consumes in the Nash equilibrium.





Exercise 2 Public Economics

Suppose that person A produces good x. The production process imposes external cost on person B. Both persons maximise their own utility, which increases in their own income. The profit that person A derives from producing q units of good x is given by $kq - \frac{1}{2}q^2$ euro. Total external cost incurred by person B are equal to cq euro. The conditions of the Coase Theorem are met and Person A has the (property) right to produce x. In the absence of government intervention, Person A and B make a socially efficient binding agreement, which determines how many units of good x person A can produce as well as a transfer of T euro from person B to person A. Assume that $k > c > 0$.

- 1p **2a** Show that the minimal transfer T that person B pays to person A equals $\frac{1}{2}c^2$.





- 1p **2b** Explain why the minimal transfer T increases in marginal external cost c . Use no more than 3 sentences.

- 1p **2c** Suppose the government maximises a social welfare function, which is equal to the sum of the utilities of persons A and B. Give one reason for why the government may decide to engage in government intervention in the situation of this exercise. Use no more than one sentence.



Exercise 3 Personnel economics

A profit-maximizing tobacco company hires a worker. The worker's utility function is: $U = w - \frac{1}{2}\theta e^2 - \beta q$, where w is the worker's income, e is the amount of effort exerted by the worker, and q is the amount of output produced by the worker. θ and β are positive parameters. The worker's income consists of a bonus b per unit of production. There is no base salary. Hence, $w = bq$. The production function is $q = \mu e$. The first term of the utility function reflects that the worker enjoys income. The second term reflects worker's cost of effort. And the third term reflects that the worker feels guilt about producing tobacco, which reduces his utility. The worker chooses to work for the tobacco company if his utility is at least equal to zero. Effort e cannot be negative.

- 2p **3a** Derive the level of effort the worker chooses to exert.

- 3p **3b** Assume each unit of production is sold by the tobacco company for price p , which is assumed sufficiently high so that the tobacco company wishes to produce a positive amount. Derive the profit-maximizing level of the bonus b .





Additional answer space

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

4a Additional answer box 1

4b Additional answer box 2

4c Additional answer box 3



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